



Lecture 5 and 6

Chemical Kinetics and Equilibria

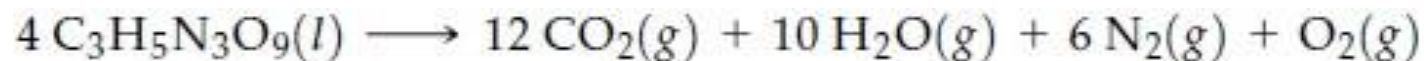
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Chemical Kinetics



Chemical reactions can be **slow, fast, or explosive**. But **explosions** are in a class by themselves. They are characterized by instantaneous release of large quantities of hot gases such as O_2 , NH_3 , CO_2 and H_2O . The gases **expand rapidly** and set up a shock wave of **enormous pressure** (up to 700,000 atm) that propagates through the surroundings at speeds of up to 9000 m/s, causing physical destruction:- witnessed in movies and television newscasts.

Because the gaseous explosion products are extremely stable, a great deal of energy is released within a few microseconds during an explosion, e.g. one mole (227 g) of nitroglycerin, **releases 1427 kJ when it explodes!!!!**





The branch of chemistry concerned with reaction rates and the sequence of steps by which reactions occur is called **chemical kinetics**.

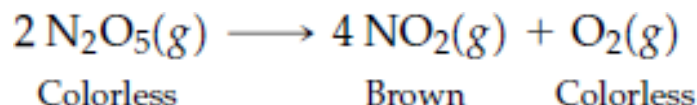


The rates of chemical reactions differ greatly. Some reactions, such as the combination of sodium and bromine, occur instantly. Other reactions, such as the rusting of iron, are slow.

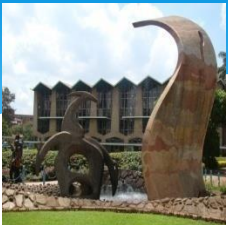
The rate of a reaction specifies how fast the concentration of a reactant or a product changes per unit time.

$$\text{Rate} = \frac{\text{Concentration change}}{\text{Time change}}$$

The thermal decomposition of gaseous dinitrogen pentoxide, to give the brown gas nitrogen dioxide and molecular oxygen:



Alternatively, concentration changes can be monitored by measuring the intensity of the brown colour NO_2 . Note that the **concentrations of NO_2 and O_2 increase as the concentration of N_2O_5 decreases.**



The **reaction rate** can be defined either as the *increase* in the concentration of a *product* per unit time or as the *decrease* in the concentration of a *reactant* per unit time.

Change in the molar concentration of oxygen

$$\text{Rate of formation of O}_2 = \frac{\Delta[\text{O}_2]}{\Delta t} = \frac{\text{Conc of O}_2 \text{ at time } t_2 - \text{Conc of O}_2 \text{ at time } t_1}{t_2 - t_1}$$

Change in time

where the square brackets surrounding O₂ denote its molar concentration;

Consider the general reaction;



where **A and B are the reactants**, and **a and b** are stoichiometric coefficients in the balanced chemical equation.



The dependence of the reaction rate on the concentration of each reactant is given by an equation called the **rate law**.



The rate law is usually written in the form;

$$\text{Rate} = - \frac{\Delta[A]}{\Delta t} = k[A]^m[B]^n$$

where k is a proportionality constant called the **rate constant**. We have arbitrarily expressed the rate as the rate of disappearance of A

The values of the exponents m and n determine the **reaction order** with respect to A and B, respectively. The sum of the exponents $m + n$ defines the *overall* reaction order. Thus, if the rate law is

$$\text{Rate} = k[A]^2[B] \quad m = 2; n = 1; \text{ and } m + n = 3$$

we say that the reaction is second order in A, first order in B, and third order overall.

The values of the exponents in a rate law must be determined by experiment; they cannot be deduced from the stoichiometry of the reaction.



Example



Initial rate data for the decomposition of gaseous N_2O_5 at 55°C are as follows:

Experiment	Initial $[\text{N}_2\text{O}_5]$	Initial Rate of Decomposition of N_2O_5 (M/s)
1	0.020	3.4×10^{-5}
2	0.050	8.5×10^{-5}

(a) What is the rate law?

(b) What is the value of the rate constant?

The rate law for the decomposition of N_2O_5 can be written as

$$\text{Rate} = - \frac{\Delta[\text{N}_2\text{O}_5]}{\Delta t} = k[\text{N}_2\text{O}_5]^m$$

where m is both the order of the reaction in N_2O_5 and the overall reaction order. To find the value of the exponent m , compare the change in the initial concentration of N_2O_5 for experiments 1 and 2 with the change in the initial rate.

(a) Comparing experiments 1 and 2 shows that an increase in the initial concentration of N_2O_5 by a factor of 2.5 increases the initial rate by a factor of 2.5:

$$\frac{[\text{N}_2\text{O}_5]_2}{[\text{N}_2\text{O}_5]_1} = \frac{0.050 \text{ M}}{0.020 \text{ M}} = 2.5 \quad \frac{(\text{Rate})_2}{(\text{Rate})_1} = \frac{8.5 \times 10^{-5} \text{ M/s}}{3.4 \times 10^{-5} \text{ M/s}} = 2.5$$

The rate depends linearly on the concentration of N_2O_5 , and therefore the rate law is

$$\text{Rate} = - \frac{\Delta[\text{N}_2\text{O}_5]}{\Delta t} = k[\text{N}_2\text{O}_5]$$

The reaction is first order in N_2O_5 . If the rate had increased by a factor of $(2.5)^2 = 6.25$, the reaction would have been second order in N_2O_5 . If the rate had increased by a factor of $(2.5)^3 = 15.6$, the reaction would have been third order in N_2O_5 ; and so on.

$$k = \frac{\text{Rate}}{[\text{N}_2\text{O}_5]} = \frac{3.4 \times 10^{-5} \frac{\text{M}}{\text{s}}}{0.020 \text{ M}} = 1.7 \times 10^{-3} \text{ s}^{-1}$$



Integrated Rate Law for a First-Order Reaction



A simple, general first-order reaction is given as:



A first-order reaction is one whose rate depends on the concentration of a single reactant raised to the first power.

$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k[A]$$

An example is the decomposition of hydrogen peroxide in basic solution:



Using calculus, it's possible to convert the rate law to another form, called the **integrated rate law**:

$$\ln \frac{[A]_t}{[A]_0} = -kt$$

$\ln$ denotes the natural logarithm, $[A]_0$ designates the concentration of A at some initial time, arbitrarily considered to be $t=0$ and $[A]_t$ is the concentration of A at any time t thereafter.



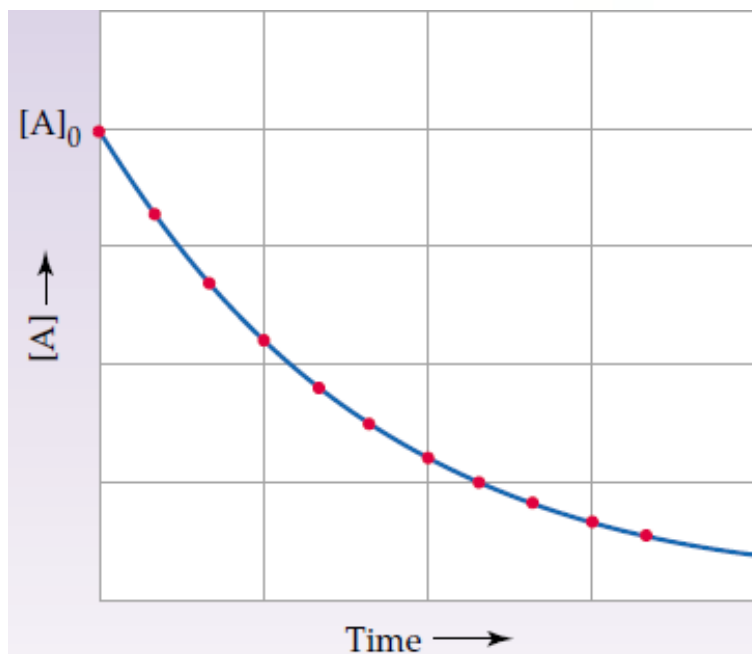
Since $\ln ([A]_t/[A]_0) = \ln [A]_t - \ln [A]_0$, we can rewrite the integrated rate law as

$$\ln [A]_t = -kt + \ln [A]_0$$

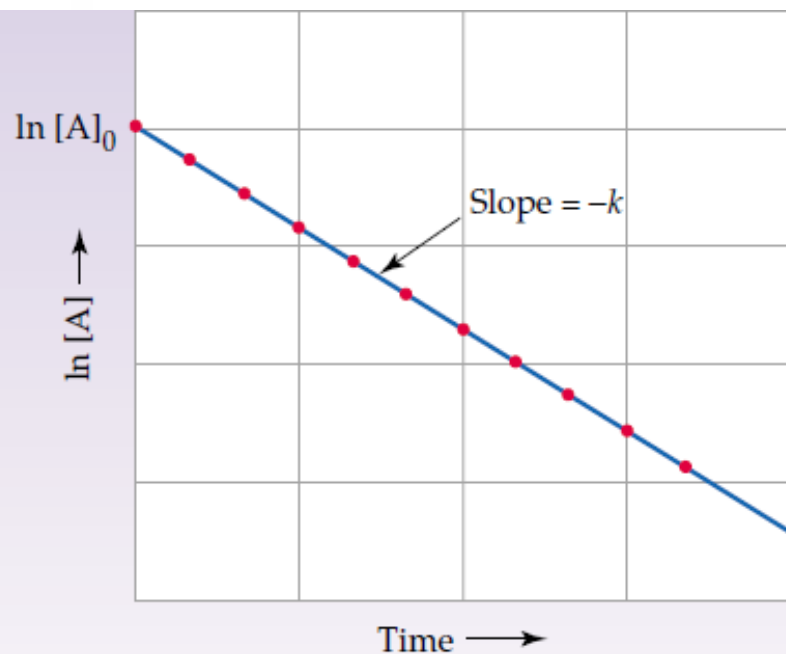
This equation is of the form $y = mx + b$, which says that $\ln [A]_t$ is a linear function of time:

$$\ln [A]_t = (-k)t + \ln [A]_0$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 y m x b



(a)



(b)



Half-Life of a First-Order Reaction



The **half-life** of a reaction, symbolized by $t_{1/2}$ is the time required for the reactant concentration to drop to one-half of its initial value.

To relate the reaction's half-life to the rate constant, let's begin with the integrated rate law:

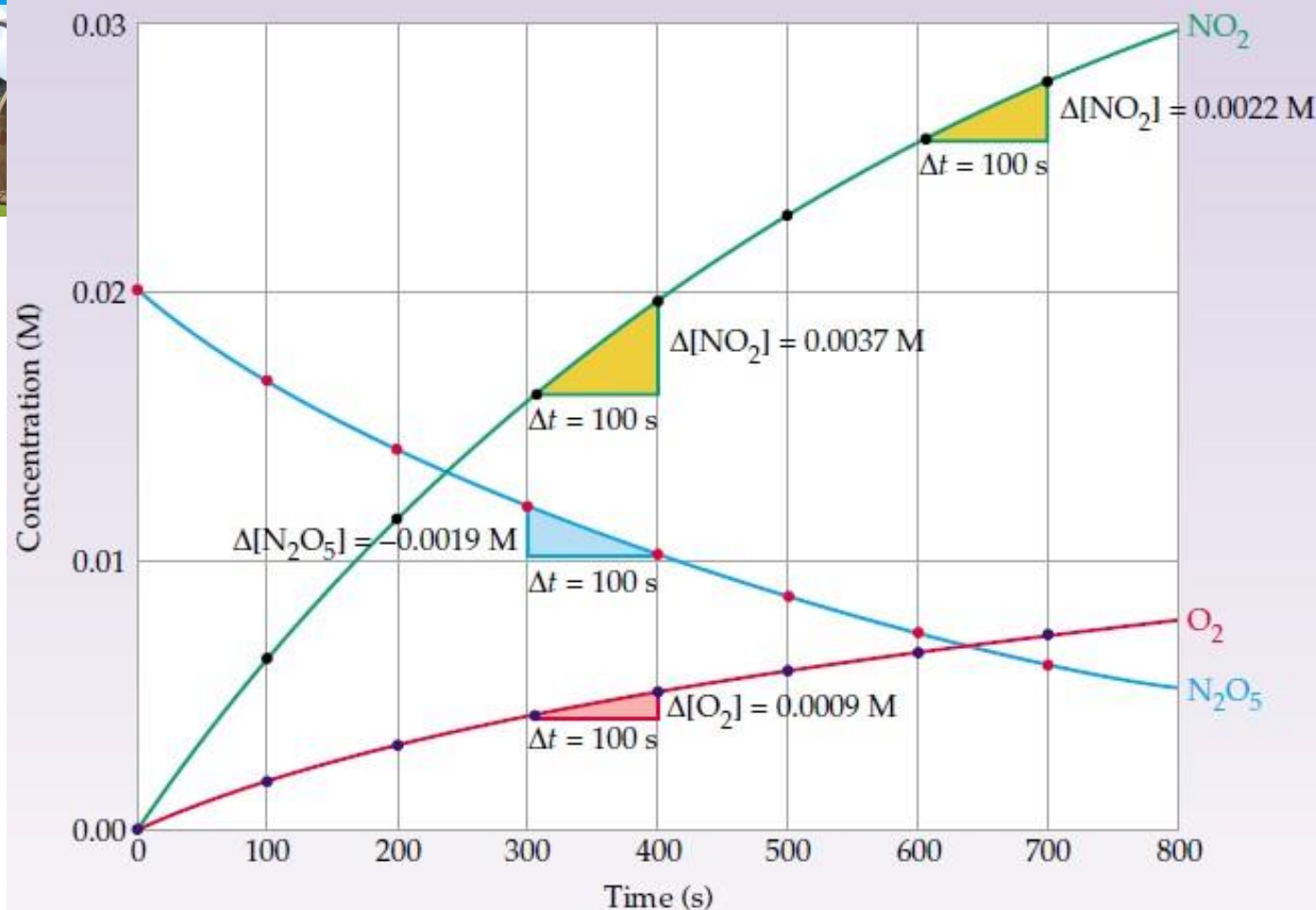
$$\ln \frac{[A]_t}{[A]_0} = -kt$$

When $t = t_{1/2}$, the fraction of A that remains, $[A]_t/[A]_0$, is 1/2. Therefore,

$$\begin{aligned}\ln \frac{1}{2} &= -kt_{1/2} \\ \text{so } t_{1/2} &= \frac{-\ln \frac{1}{2}}{k} = \frac{\ln 2}{k} \\ \text{or } t_{1/2} &= \frac{0.693}{k}\end{aligned}$$

Thus, the half-life of a first-order reaction is readily calculated from the rate constant, and vice versa.

Example



- Estimate the half-life for the decomposition of gaseous N_2O_5 at 55°C from the concentration-versus-time plot in Figure
- Calculate the half-life from the rate constant ($1.7 \times 10^{-3} \text{ s}^{-1}$).
- If the initial concentration of N_2O_5 is 0.020 M, what is the concentration of N_2O_5 after five half-lives?
- How long will it take for the N_2O_5 concentration to fall to 12.5% of its initial value?



SOLUTION

(a) Figure shows that the conc. of N_2O_5 falls from 0.020 M to 0.010 M during a time period of approximately 400 s. At 800 s, has decreased by another factor of 2, to 0.0050 M. Therefore, $t_{1/2} = 400$ s

(b) Based on the value of the rate constant,

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{1.7 \times 10^{-3} \text{ s}^{-1}} = 4.1 \times 10^2 \text{ s (6.8 min)}$$

(c) At $5t_{1/2}$, $[\text{N}_2\text{O}_5]$ will be $(1/2)^5 = 1/32$ of its initial value. Therefore,

$$[\text{N}_2\text{O}_5] = \frac{0.020 \text{ M}}{32} = 0.000 62 \text{ M}$$

(d) Since 12.5% of the initial concentration corresponds to $1/8$ or $(1/2)^3$ of the initial concentration, the time required is three half-lives:

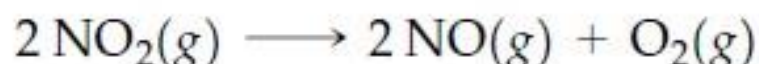
$$t = 3t_{1/2} = 3(4.1 \times 10^2 \text{ s}) = 1.2 \times 10^3 \text{ s (20 min)}$$

Second-Order Reactions

A second-order reaction is one whose rate depends on the concentration of a single reactant raised to the second power or on the concentrations of two different reactants, each raised to the first power. For the simpler type, $A \rightarrow \text{products}$, the rate law is

$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k[A]^2$$

An example is the thermal decomposition of nitrogen dioxide to yield NO and O₂:



Using calculus, it's possible to convert the rate law to the integrated rate law

$$\underset{\substack{\uparrow \\ y}}{\frac{1}{[A]_t}} = \underset{\substack{\uparrow \uparrow \\ mx}}{kt} + \underset{\substack{\uparrow \\ b}}{\frac{1}{[A]_0}}$$

This integrated rate law then allows us to calculate the concentration of A at any time t if the initial concentration $[A]_0$ is known.

The slope of the straight line is the rate constant k , and the intercept is $1/[A]_0$. Thus, by plotting $1/[A]$ versus time, we can test whether the reaction is second order and can determine the value of the rate constant



Half-life of a second-order reaction



$$\frac{1}{\left(\frac{[A]_0}{2}\right)} = kt_{1/2} + \frac{1}{[A]_0}$$

$$\text{so } t_{1/2} = \frac{1}{k} \left(\frac{2}{[A]_0} - \frac{1}{[A]_0} \right)$$

$$\text{or } t_{1/2} = \frac{1}{k[A]_0}$$

Comparative analysis

	First-Order	Second-Order
Rate law	$-\frac{\Delta[A]}{\Delta t} = k[A]$	$-\frac{\Delta[A]}{\Delta t} = k[A]^2$
Concentration-time equation	$\ln [A]_t = -kt + \ln [A]_0$	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$
Linear graph	$\ln [A]$ versus t	$\frac{1}{[A]}$ versus t
Graphical determination of k	$k = -(\text{Slope})$	$k = \text{Slope}$
Half-life	$t_{1/2} = \frac{0.693}{k}$ (constant)	$t_{1/2} = \frac{1}{k[A]_0}$ (not constant)



Example



At elevated temperatures, nitrogen dioxide decomposes to nitric oxide and molecular oxygen:



Concentration–time data for consumption of NO_2 at 300°C are as follows:

Time (s)	$[\text{NO}_2]$	Time (s)	$[\text{NO}_2]$
0	8.00×10^{-3}	200	4.29×10^{-3}
50	6.58×10^{-3}	300	3.48×10^{-3}
100	5.59×10^{-3}	400	2.93×10^{-3}
150	4.85×10^{-3}	500	2.53×10^{-3}

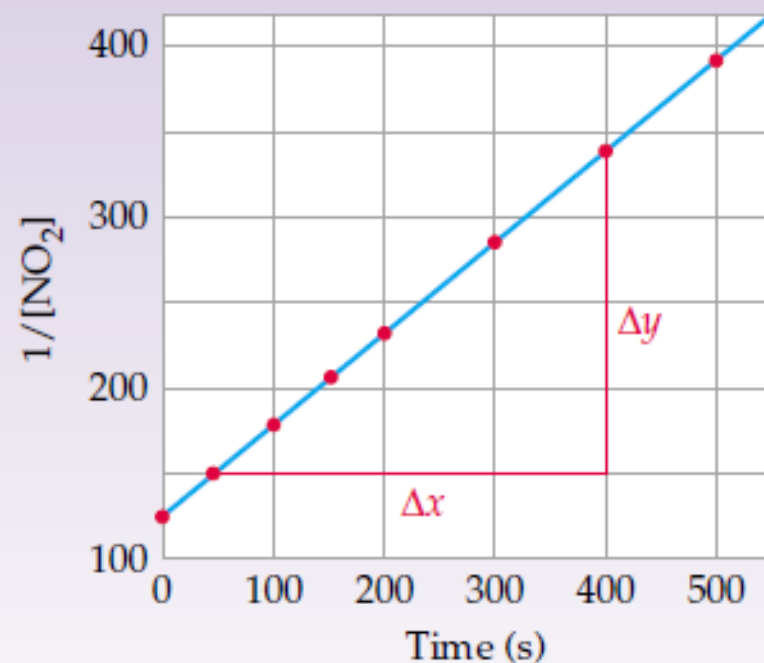
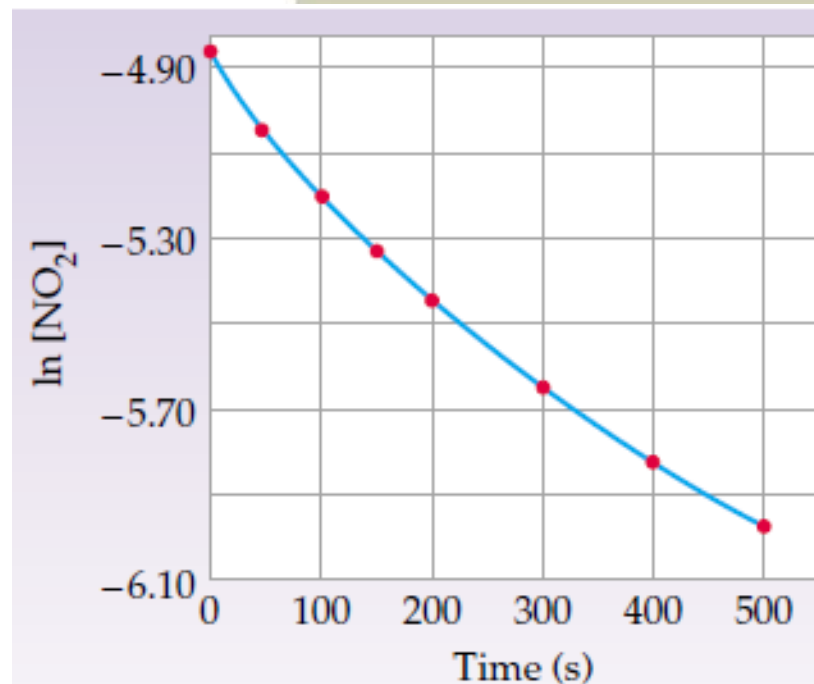
- (a) Is the reaction first order or second order?
- (b) What is the value of the rate constant?
- (c) What is the concentration of NO_2 at $t = 20.0 \text{ min}$?
- (d) What is the half-life of the reaction when the initial concentration of NO_2 is $6.00 \times 10^{-3} \text{ M}$?
- (e) What is $t_{1/2}$ when $[\text{NO}_2]_0$ is $3.00 \times 10^{-3} \text{ M}$?



(a) The plot of $\ln [\text{NO}_2]$ versus time is curved, but the plot of $1/[\text{NO}_2]$ versus time is a straight line. The reaction is therefore second order in NO_2 .



Time (s)	$[\text{NO}_2]$	$\ln [\text{NO}_2]$	$1/[\text{NO}_2]$
0	8.00×10^{-3}	-4.828	125
50	6.58×10^{-3}	-5.024	152
100	5.59×10^{-3}	-5.187	179
150	4.85×10^{-3}	-5.329	206
200	4.29×10^{-3}	-5.451	233
300	3.48×10^{-3}	-5.661	287
400	2.93×10^{-3}	-5.833	341
500	2.53×10^{-3}	-5.980	395



(b) The rate constant equals the slope of the straight line in the plot of $1/[\text{NO}_2]$ versus time, which we can estimate from the coordinates of two widely separated points on the line:

$$k = \text{Slope} = \frac{\Delta y}{\Delta x} = \frac{340 \text{ M}^{-1} - 150 \text{ M}^{-1}}{400 \text{ s} - 50 \text{ s}} = \frac{190 \text{ M}^{-1}}{350 \text{ s}} = 0.54/(\text{M} \cdot \text{s})$$

(c) The concentration of NO_2 at $t = 20.0 \text{ min}$ ($1.20 \times 10^3 \text{ s}$) can be calculated using the integrated rate law:

$$\frac{1}{[\text{NO}_2]_t} = kt + \frac{1}{[\text{NO}_2]_0}$$

Substituting the values of k , t , and $[\text{NO}_2]_0$ gives

$$\begin{aligned} \frac{1}{[\text{NO}_2]_t} &= \left(\frac{0.54}{\text{M} \cdot \text{s}} \right) (1.20 \times 10^3 \text{ s}) + \frac{1}{8.00 \times 10^{-3} \text{ M}} \\ &= \frac{648}{\text{M}} + \frac{125}{\text{M}} = \frac{773}{\text{M}} \end{aligned}$$

$$[\text{NO}_2]_t = 1.3 \times 10^{-3} \text{ M}$$

(d) The half-life of a second-order reaction can be calculated from the rate constant and the initial concentration of NO_2 ($6.00 \times 10^{-3} \text{ M}$):

$$t_{1/2} = \frac{1}{k[\text{NO}_2]_0} = \frac{1}{\left(\frac{0.54}{\text{M} \cdot \text{s}} \right) (6.00 \times 10^{-3} \text{ M})} = 3.1 \times 10^2 \text{ s}$$

(e) When $[\text{NO}_2]_0$ is $3.00 \times 10^{-3} \text{ M}$, $t_{1/2} = 6.2 \times 10^2 \text{ s}$ (twice as long as when $[\text{NO}_2]_0$ is $6.00 \times 10^{-3} \text{ M}$ because $[\text{NO}_2]_0$ is now smaller by a factor of 2).



Reaction Rates and Temperature: The Arrhenius Equation



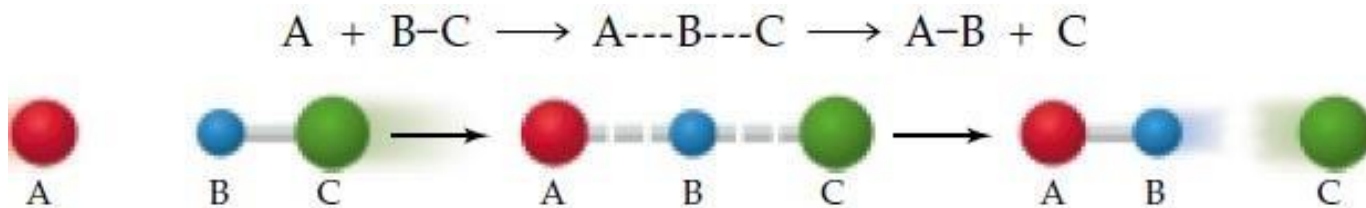
According to the **collision theory** model, a bimolecular reaction occurs when two properly oriented reactant molecules come together in a sufficiently energetic collision. To be specific, let's consider one of the simplest possible reactions, the reaction of an atom A with a diatomic molecule BC to give a diatomic molecule AB and an atom C:



An example from atmospheric chemistry is the reaction of an oxygen atom with an HCl molecule to give an OH molecule and a chlorine atom:



Between the reactant and product stages, the nuclei pass through a configuration in which all three atoms are weakly linked together. We can picture the progress of the reaction as;

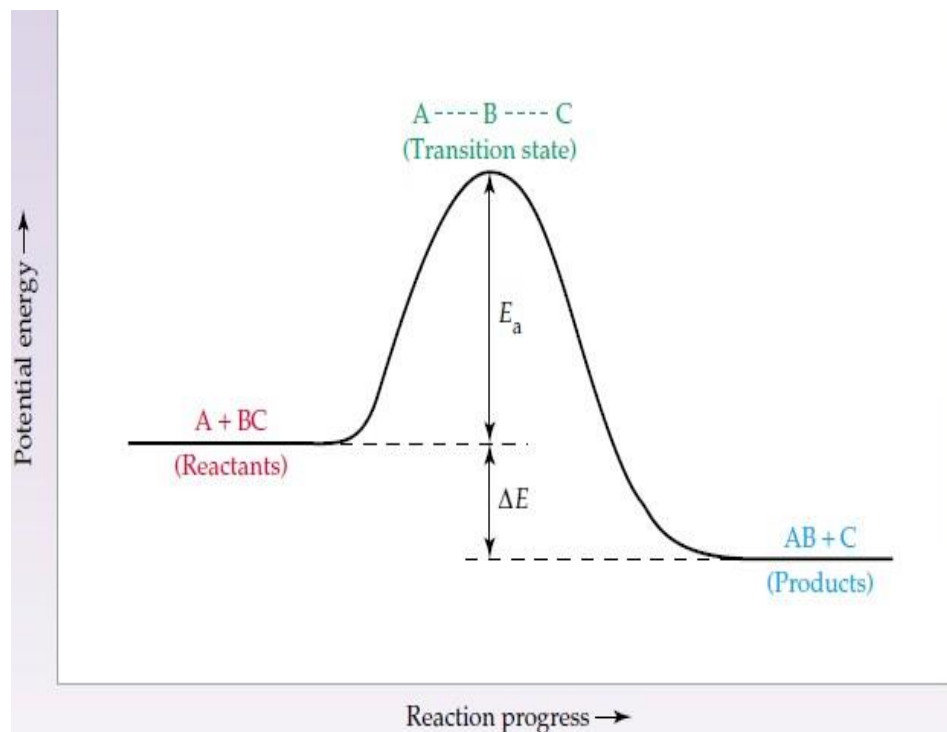


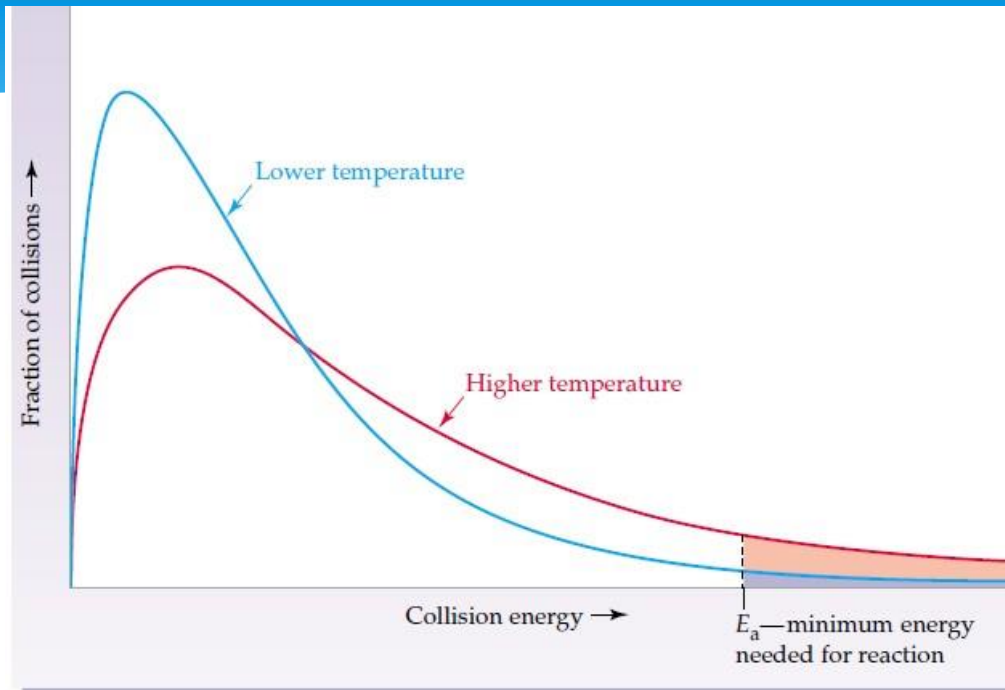


A---B---C has more potential energy than either the reactants or the products. Thus, there is a potential energy barrier that must be surmounted before reactants can be converted to products, as depicted graphically



The height of the barrier is called the **activation energy**, E_a and the configuration of atoms at the maximum in the potential energy profile is called the **transition state**, or the *activated complex*. Since energy is conserved in the collision, all the energy needed to climb the potential energy hill must come from the **kinetic energy of the colliding molecules**. If the collision energy is less than E_a the reactant molecules can't surmount the barrier and they simply bounce apart. If the collision energy is at least as great as E_a , however, the reactants can climb over the barrier and be converted to products.





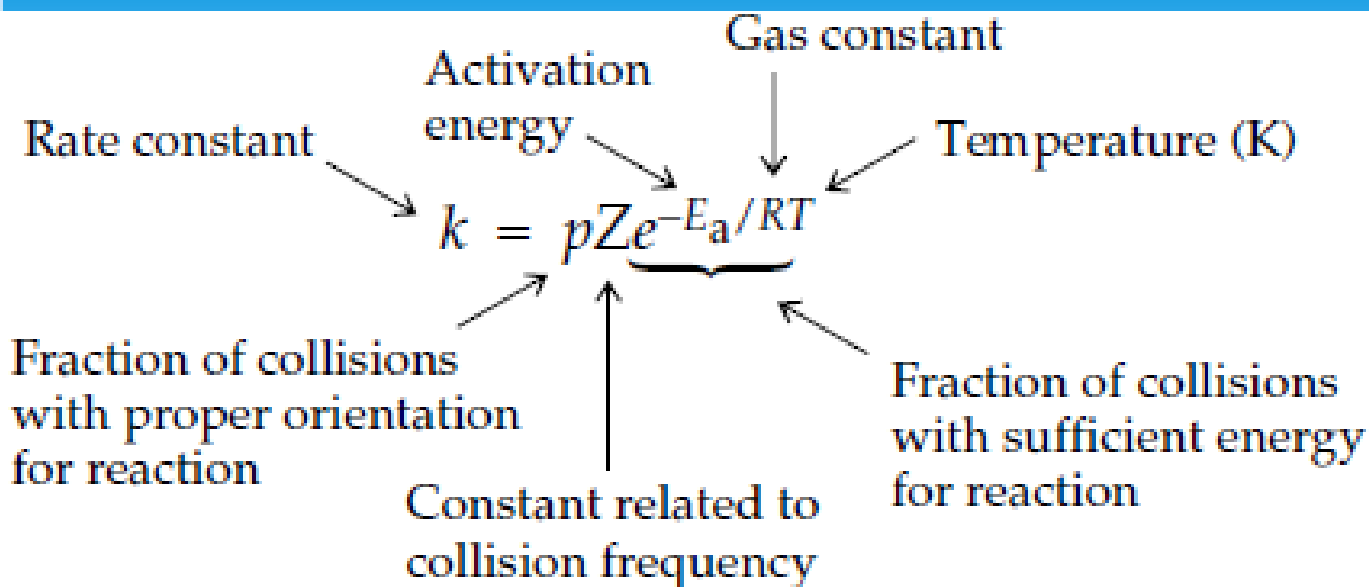
The reaction rate is lower than the collision rate by a factor $p \times f$ because only a fraction of the colliding molecules have the correct orientation and the minimum energy needed for reaction:

$$\text{Reaction rate} = p \times f \times \text{Collision rate} = pfZ[A][B]$$

Since the rate law is

$$\text{Reaction rate} = k[A][B]$$

the rate constant predicted by collision theory is $k = pfZ$, or $k = pZe^{-E_a/RT}$.



This expression is usually written in a form called the **Arrhenius equation**;

$$k = Ae^{-E_a/RT}$$

The parameter A ($= pZ$) is called the **frequency factor** (or pre-exponential factor). The minus sign in the exponent, indicates that, the rate constant decreases as T increases.



Using the Arrhenius Equation



Taking the natural logarithm of both sides of the Arrhenius equation, we obtain the logarithmic form;

$$\ln k = \ln A - \frac{E_a}{RT}$$

This equation can be rearranged into the form $y = mx + b$, so a graph of $\ln k$ versus $1/T$ (called an Arrhenius plot) gives a straight line with slope $m = -E_a/R$ and intercept $b = \ln A$:

$$\begin{array}{ccccccc} \ln k & = & \left(\frac{-E_a}{R} \right) & \left(\frac{1}{T} \right) & + & \ln A \\ \uparrow & & \uparrow & \uparrow & & \uparrow \\ y & & m & x & & b \end{array}$$

The experimental value of the activation energy can be determined from the slope of the straight line, as shown in

$$E_a = -R(\text{Slope})$$



Another form of the Arrhenius equation can be derived that allows us to estimate the activation energy from rate constants at just two temperatures. At temperatures T_1 and T_2 ;



$$\ln k_1 = \left(\frac{-E_a}{R} \right) \left(\frac{1}{T_1} \right) + \ln A \quad \ln k_2 = \left(\frac{-E_a}{R} \right) \left(\frac{1}{T_2} \right) + \ln A$$

$$\ln \left(\frac{k_2}{k_1} \right) = \left(\frac{-E_a}{R} \right) \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Example

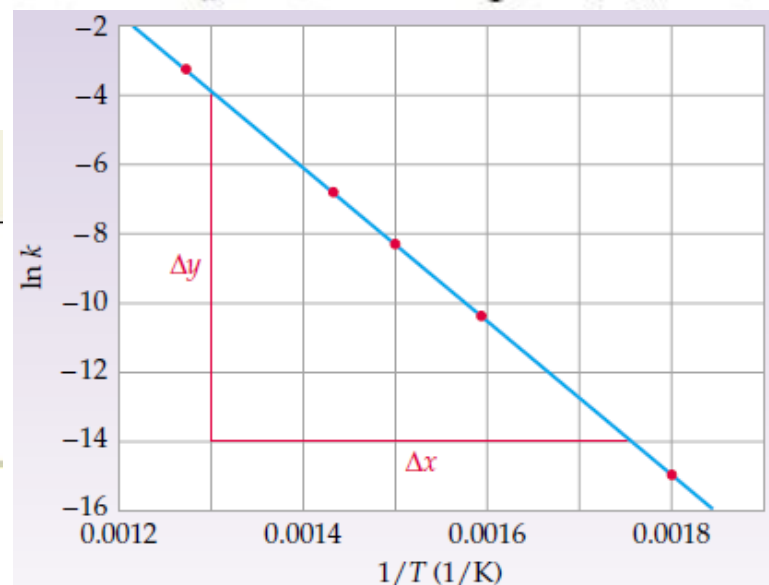
Rate constants for the gas-phase decomposition of hydrogen iodide, $2 \text{HI}(g) \rightarrow \text{H}_2(g) + \text{I}_2(g)$, are listed in the following table:

Temperature (°C)	k ($\text{M}^{-1} \text{s}^{-1}$)	Temperature (°C)	k ($\text{M}^{-1} \text{s}^{-1}$)
283	3.52×10^{-7}	427	1.16×10^{-3}
356	3.02×10^{-5}	508	3.95×10^{-2}
393	2.19×10^{-4}		

- Find the activation energy (in kJ/mol) using all five data points.
- Calculate E_a from the rate constants at 283°C and 508°C.
- Given the rate constant at 283°C and the value of E_a obtained in part (b), what is the rate constant at 293°C?

Solution

t (°C)	T (K)	k (M ⁻¹ s ⁻¹)	$1/T$ (1/K)	$\ln k$
283	556	3.52×10^{-7}	0.001 80	-14.860
356	629	3.02×10^{-5}	0.001 59	-10.408
393	666	2.19×10^{-4}	0.001 50	-8.426
427	700	1.16×10^{-3}	0.001 43	-6.759
508	781	3.95×10^{-2}	0.001 28	-3.231



The slope of the straight-line plot can be determined from the coordinates of two widely separated points:

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{(-14.0) - (-3.9)}{(0.001\,75\,\text{K}^{-1}) - (0.001\,30\,\text{K}^{-1})} = \frac{-10.1}{0.000\,45\,\text{K}^{-1}} = -2.24 \times 10^4\,\text{K}$$

Finally, calculate the activation energy from the slope:

$$\begin{aligned} E_a &= -R(\text{Slope}) = -\left(8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}}\right)(-2.24 \times 10^4\,\text{K}) \\ &= 1.9 \times 10^5\,\text{J/mol} = 190\,\text{kJ/mol} \end{aligned}$$



(b) Substituting the values of $k_1 = 3.52 \times 10^{-7} \text{ M}^{-1} \text{ s}^{-1}$ at $T_1 = 556 \text{ K}$ (283°C) and $k_2 = 3.95 \times 10^{-2} \text{ M}^{-1} \text{ s}^{-1}$ at $T_2 = 781 \text{ K}$ (508°C) into the equation

$$\ln \left(\frac{k_2}{k_1} \right) = \left(\frac{-E_a}{R} \right) \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

gives

$$\ln \left(\frac{3.95 \times 10^{-2} \text{ M}^{-1} \text{ s}^{-1}}{3.52 \times 10^{-7} \text{ M}^{-1} \text{ s}^{-1}} \right) = \left(\frac{-E_a}{8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}}} \right) \left(\frac{1}{781 \text{ K}} - \frac{1}{556 \text{ K}} \right)$$

and simplifying this equation gives

$$11.628 = \left(\frac{-E_a}{8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}}} \right) \left(\frac{-5.18 \times 10^{-4}}{\text{K}} \right)$$

$$E_a = 1.87 \times 10^5 \text{ J/mol} = 187 \text{ kJ/mol}$$



(c) Use the same equation as in part (b), but now the known values are $k_1 = 3.52 \times 10^{-7} \text{ M}^{-1} \text{ s}^{-1}$ at $T_1 = 556 \text{ K}$ (283°C) and $E_a = 1.87 \times 10^5 \text{ J/mol}$, and k_2 at $T_2 = 566 \text{ K}$ (293°C) is the unknown:

$$\ln \left(\frac{k_2}{3.52 \times 10^{-7} \text{ M}^{-1} \text{ s}^{-1}} \right) = \left(\frac{-1.87 \times 10^5 \frac{\text{J}}{\text{mol}}}{8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}}} \right) \left(\frac{1}{566 \text{ K}} - \frac{1}{556 \text{ K}} \right) = 0.715$$

Taking the antiln of both sides gives

$$\frac{k_2}{3.52 \times 10^{-7} \text{ M}^{-1} \text{ s}^{-1}} = e^{0.715} = 2.04$$
$$k_2 = 7.18 \times 10^{-7} \text{ M}^{-1} \text{ s}^{-1}$$

In this temperature range, a rise in temperature of 10 K doubles the rate constant.



Lecture 6

Chemical Equilibrium



Outline

- ✓ The Equilibrium State
- ✓ The Equilibrium Constant
- ✓ The Equilibrium Constant
- ✓ Heterogeneous Equilibria
- ✓ Using the Equilibrium Constant
- ✓ Factors That Alter the Composition of an Equilibrium Mixture



Chemical Equilibrium



Two important questions about chemical reactions:

How fast does a reaction happen?

The kinetics of the reaction

To what extent does it happen?

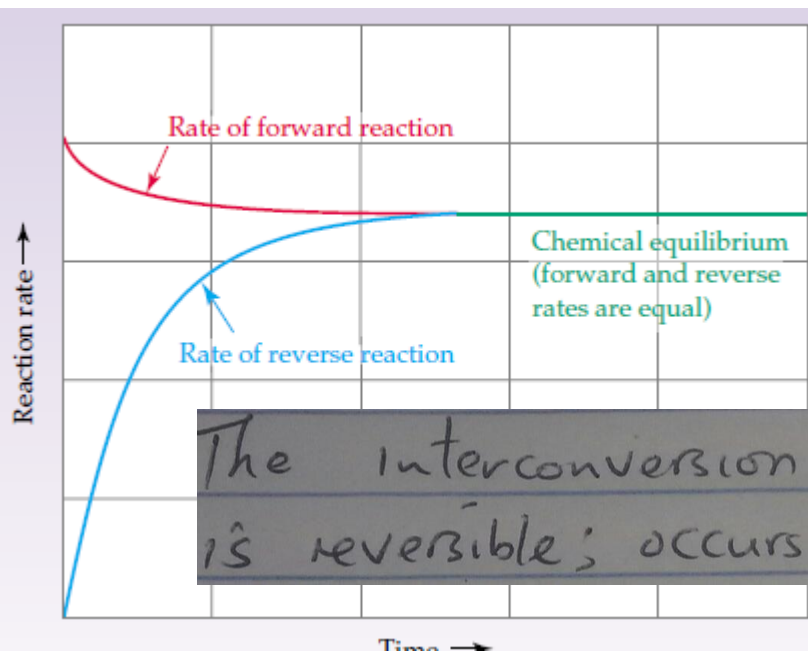
This lecture

CHEMICAL EQUILIBRIUM

The state reached when the concentrations of reactants and products remain constant over time

For example

All reactions are reversible, irreversible reaction may as well mean reactions that proceed to completion; the reverse reaction becoming too slow to be detected.



The interconversion of $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ is reversible; occurs in both directions

The Equilibrium Constants



For the equilibrium reaction;



Equilibrium equation: $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$

← Products
← Reactants

Equilibrium constant

Equilibrium constant expression

$$c C + d D \rightleftharpoons a A + b B \quad K_c' = \frac{[A]^a[B]^b}{[C]^c[D]^d} = \frac{1}{K_c}$$

Equilibrium Constant, K_p

Equilibrium equations for gas-phase reactions are often written using partial pressures rather than molar concentrations. For example, the equilibrium equation for the decomposition of N_2O_4 can be written as for the reaction;

$$K_p = \frac{(P_{NO_2})^2}{P_{N_2O_4}} \quad \text{for the reaction } N_2O_4(g) \rightleftharpoons 2 NO_2(g)$$



The constants K_c and K_p for the general gas-phase reaction $a A + b B \rightleftharpoons c C + d D$ are related because the pressure of each component in a mixture of ideal gases is directly proportional to its molar concentration. For component A, for example,

$$P_A V = n_A R T$$

so
$$P_A = \frac{n_A}{V} R T = [A] R T$$

Similarly, $P_B = [B] R T$, $P_C = [C] R T$, and $P_D = [D] R T$. The equilibrium equation for K_p is therefore given by

$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b} = \frac{([C] R T)^c ([D] R T)^d}{([A] R T)^a ([B] R T)^b} = \frac{[C]^c [D]^d}{[A]^a [B]^b} \times (R T)^{(c+d)-(a+b)}$$

Because the first term on the right side equals K_c , the values of K_p and K_c are related by the equation

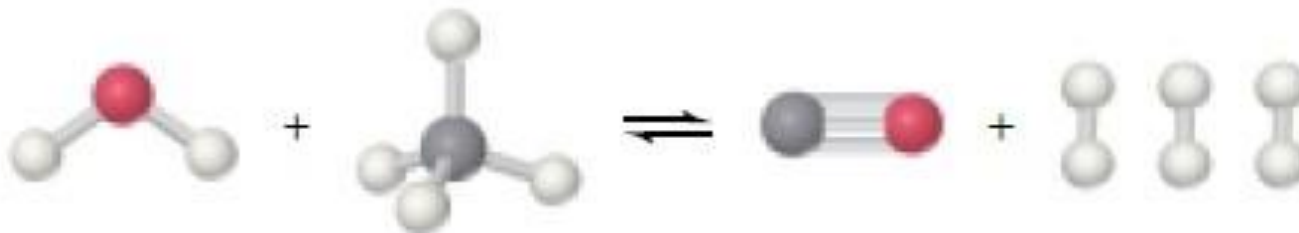
$$K_p = K_c (R T)^{\Delta n} \rightarrow \text{is the number of moles of gaseous products minus the number of moles of gaseous reactants.}$$



Examples



Hydrogen is produced industrially by the steam–hydrocarbon re-forming process. The reaction that takes place in the first step of this process is



- (a) If $K_c = 3.8 \times 10^{-3}$ at 1000 K, what is the value of K_p at the same temperature?
- (b) If $K_p = 6.1 \times 10^4$ at 1125°C, what is the value of K_c at 1125°C?



(a) For this reaction, $\Delta n = (1 + 3) - (1 + 1) = 2$. Therefore,

$$K_p = K_c(RT)^{\Delta n} = K_c(RT)^2 = (3.8 \times 10^{-3})[(0.082\ 06)(1000)]^2 = 26$$

(b) Solving the equation $K_p = K_c(RT)^2$ for K_c gives

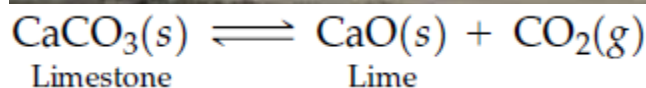
$$K_c = \frac{K_p}{(RT)^2} = \frac{6.1 \times 10^4}{[(0.082\ 06)(1398)]^2} = 4.6$$



Heterogeneous Equilibria

Reactants and products are present in more than one phase.

For example, the thermal decomposition of solid calcium carbonate, a reaction used in manufacturing cement;



$$"K_c" = \frac{[\text{CaO}][\text{CO}_2]}{[\text{CaCO}_3]}$$

$$"K_c" \times \frac{[\text{CaCO}_3]}{[\text{CaO}]} = [\text{CO}_2]$$

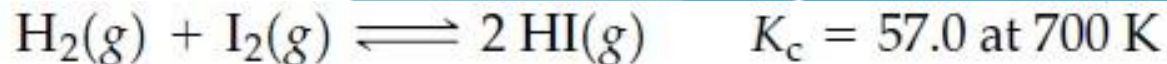
Constant Constant

As a general rule, the concentrations of pure solids and pure liquids are not included when writing an equilibrium equation because their concentrations are constants that are incorporated into the value of the equilibrium constant.

Predicting the Direction of Reaction



Consider

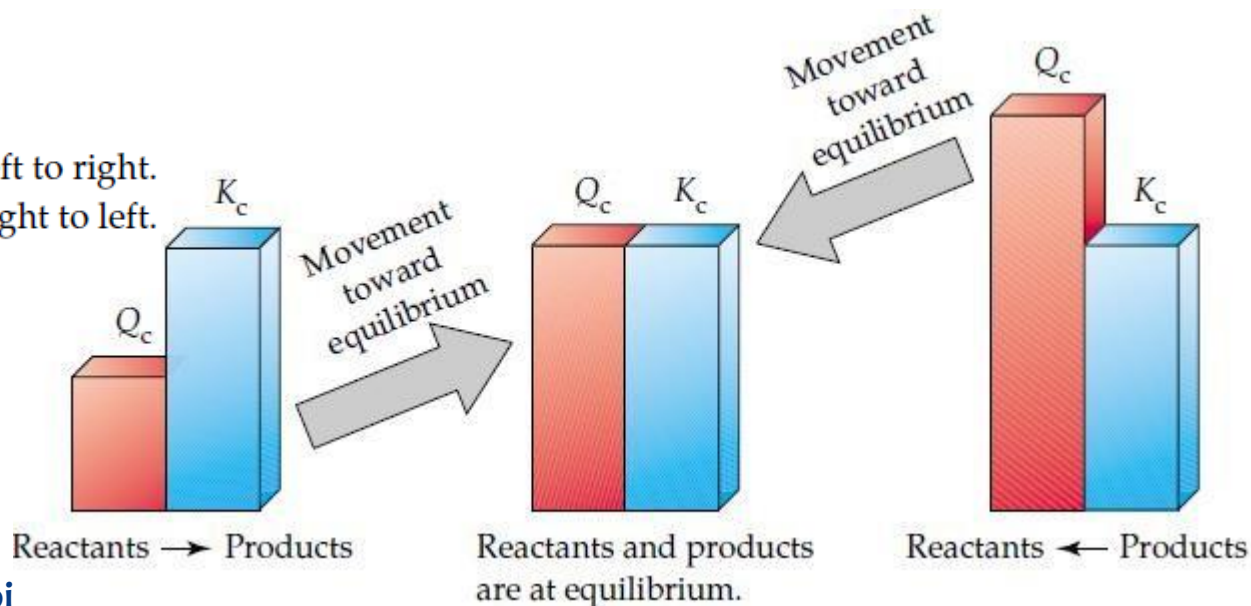


Suppose that we have a mixture of $\text{H}_2(\text{g})$, $\text{I}_2(\text{g})$, and $\text{HI}(\text{g})$ at 700 K with concentrations $[\text{H}_2]_t = 0.10 \text{ M}$, $[\text{I}_2]_t = 0.20 \text{ M}$, and $[\text{HI}]_t = 0.40 \text{ M}$.

*The subscript t on the concentration symbols means that the concentrations were measured at some arbitrary time t , not necessarily at equilibrium.) If we substitute these concentrations into the equilibrium constant expression, we obtain a value called the **reaction quotient***

Reaction quotient:
$$Q_c = \frac{[\text{HI}]_t^2}{[\text{H}_2]_t[\text{I}_2]_t} = \frac{(0.40)^2}{(0.10)(0.20)} = 8.0$$

- If $Q_c < K_c$, net reaction goes from left to right.
- If $Q_c > K_c$, net reaction goes from right to left.
- If $Q_c = K_c$, no net reaction occurs.



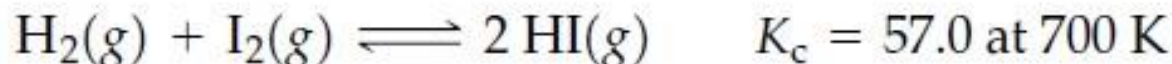


More examples



Take a scenario where we know the initial concentrations but do not know any of the equilibrium concentrations.

The equilibrium constant K_c for the reaction of H_2 with I_2 is 57.0 at 700 K:



If 1.00 mol of H_2 is allowed to react with 1.00 mol of I_2 in a 10.0 L reaction vessel at 700 K, what are the concentrations of H_2 , I_2 , and HI at equilibrium? What is the composition of the equilibrium mixture in moles?

Solution	$H_2(g)$	+	$I_2(g)$	$\rightleftharpoons$	$2 HI(g)$
Initial concentration (M)	0.100		0.100		0
Change (M)	$-x$		$-x$		$+2x$
Equilibrium concentration (M)	$(0.100 - x)$		$(0.100 - x)$		$2x$

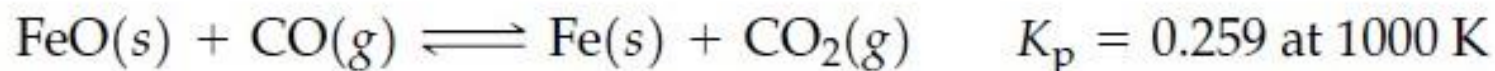
$$K_c = 57.0 = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2x)^2}{(0.100 - x)(0.100 - x)} = \left(\frac{2x}{0.100 - x} \right)^2$$

Solving $x = 0.0791$ or 0.136 M. Because the initial concentrations of and are 0.100 M, x can't exceed 0.100 M. Therefore, discard 0.136 as chemically unreasonable. Calculate the equilibrium concentrations from the calculated value of x : Check the results by substituting them into the equilibrium equation:



Calculate the equilibrium concentrations of H_2 , I_2 , and HI at 700 K if the initial concentrations are $[\text{H}_2] = 0.100 \text{ M}$ and $[\text{I}_2] = 0.200 \text{ M}$. The equilibrium constant K_c for the reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2 \text{HI}(\text{g})$ is 57.0 at 700 K.

One reaction that occurs in producing steel from iron ore is the reduction of iron(II) oxide by carbon monoxide to give iron metal and carbon dioxide. The equilibrium constant K_p for the reaction at 1000 K is 0.259.



What are the equilibrium partial pressures of CO and CO_2 at 1000 K if the initial partial pressures are $P_{\text{CO}} = 1.000 \text{ atm}$ and $P_{\text{CO}_2} = 0.500 \text{ atm}$?

	FeO(s)	+	CO(g)	$\rightleftharpoons$	Fe(s)	+	CO₂(g)
Initial pressure (atm)			1.000				0.500
Change (atm)			-x				-x
Equilibrium pressure (atm)			(1.000 - x)				(0.500 - x)

$$K_p = 0.259 = \frac{P_{\text{CO}_2}}{P_{\text{CO}}} = \frac{0.500 - x}{1.000 - x}$$



Factors That Alter the Composition of an Equilibrium Mixture

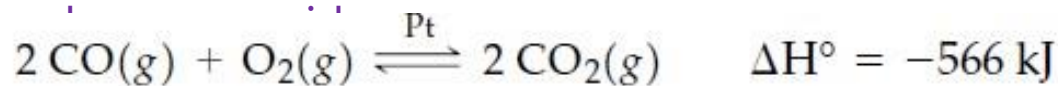


LE CHÂTELIER'S PRINCIPLE

If a stress is applied to a reaction mixture at equilibrium, net reaction occurs in the direction that relieves the stress.

Questions

1. Discuss the Factors That Alter the Composition of an Equilibrium Mixture
2. A platinum catalyst is used in automobile catalytic converters to hasten the



Suppose that you have a reaction vessel containing an equilibrium mixture of $\text{CO}(g)$, $\text{O}_2(g)$, and $\text{CO}_2(g)$. Will the amount of CO increase, decrease, or remain the same when:

- (a) A platinum catalyst is added
- (b) The temperature is increased
- (c) The pressure is increased by decreasing the volume
- (d) The pressure is increased by adding argon gas
- (e) The pressure is increased by adding O_2 gas

Solubility Equilibria

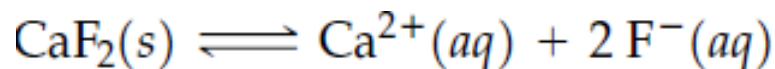


Many biological and environmental processes involve the dissolution or precipitation of a sparingly soluble ionic compound.

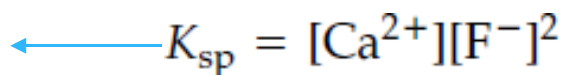


Tooth decay, for example, begins when tooth enamel, composed of the mineral hydroxyapatite, $\text{Ca}_5(\text{PO}_4)_3\text{OH}$.

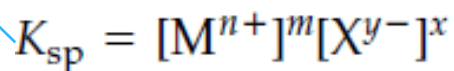
Consider the solubility equilibrium in a saturated solution of calcium fluoride in contact with an excess of solid calcium fluoride.



Solubility product constant, or simply the **solubility product**



Generally;





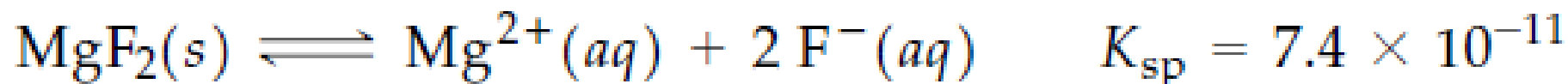
Example



Calculate the solubility of MgF_2 in water at 25°C in units of:

(a) Moles per liter

(b) Grams per liter



$$K_{\text{sp}} = 7.4 \times 10^{-11} = [\text{Mg}^{2+}][\text{F}^-]^2 = (x)(2x)^2$$

$$4x^3 = 7.4 \times 10^{-11}$$

$$x^3 = 1.8 \times 10^{-11}$$

$$x = [\text{Mg}^{2+}] = \text{Molar solubility} = 2.6 \times 10^{-4} \text{ mol/L}$$

$$\text{Solubility (in g/L)} = \frac{2.6 \times 10^{-4} \text{ mol}}{\text{L}} \times \frac{62.3 \text{ g}}{\text{mol}} = 1.6 \times 10^{-2} \text{ g/L}$$



Factors Affecting Solubility

- a) Common ion effect,
- b) Formation of complex ions,
- c) pH



1. The Common-Ion Effect (consider the MF_2 example)

If this increases ... this must decrease ... because this is constant.

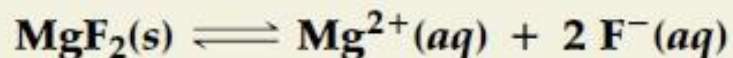


If this increases ... this must decrease ... because this is constant.

Example

Calculate the molar solubility of MgF_2 in 0.10 M NaF at 25°C.

Solubility equilibrium:



Equilibrium conc (M)

x

$0.10 + 2x$

Substituting the equilibrium concentrations into the expression for K_{sp} gives

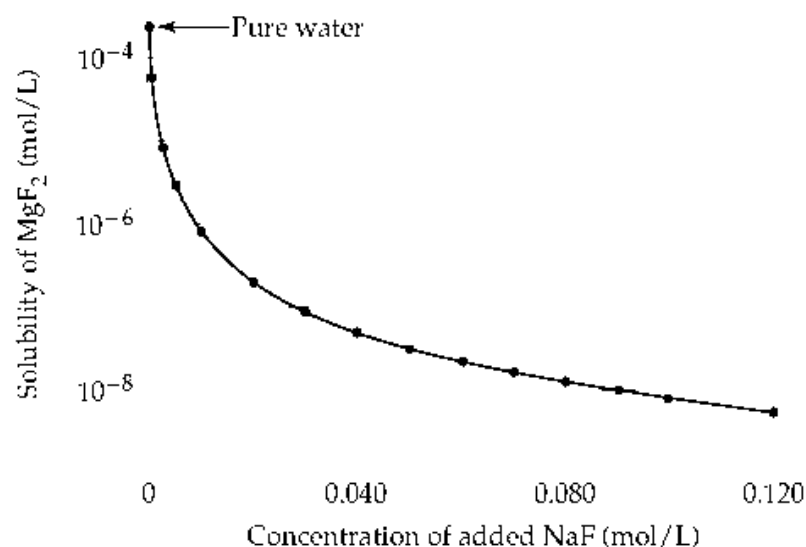
$$K_{sp} = 7.4 \times 10^{-11} = [\text{Mg}^{2+}][\text{F}^-]^2 = (x)(0.10 + 2x)^2$$

Because K_{sp} is small, $2x$ will be small compared to 0.10, and we can make the approximation that $(0.10 + 2x) \approx 0.10$. Therefore,

$$7.4 \times 10^{-11} = (x)(0.10 + 2x)^2 \approx (x)(0.10)^2$$

$$x = [\text{Mg}^{2+}] = \text{Molar solubility} = \frac{7.4 \times 10^{-11}}{(0.10)^2} = 7.4 \times 10^{-9} \text{ M}$$

Note that the calculated solubility of MgF_2 in 0.10 M NaF is less than that in pure water by a factor of about 35,000!



The common-ion effect. The solubility of MgF_2 at 25°C decreases markedly on addition of F^- ions.

Dr Bosire Geoffrey, University of Nairobi

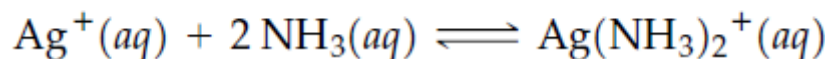
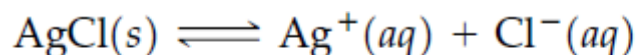


2. Formation of complex ions

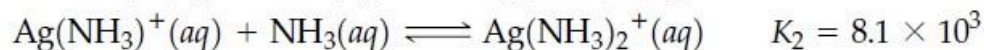
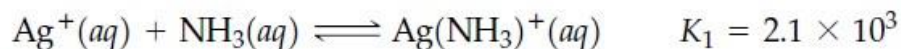


The solubility of an ionic compound increases dramatically if the solution contains a Lewis base that can form a coordinate covalent bond to the metal cation.

- Silver chloride, for example, is insoluble in water and in acid, but it dissolves in an excess of aqueous ammonia, forming the *complex ion* $[Ag(NH_3)_2]^+$
- A complex ion contains a metal cation bonded to one or more small molecules or ions, such as NH_3 , CN^- , or OH^-**
- According to Le Châtelier's principle, ammonia shifts the solubility equilibrium to the right by tying up the Ag^+ ion in the form of the complex ion:



The formation of a complex ion is a stepwise process, and each step has its own characteristic equilibrium constant. For formation of $Ag(NH_3)_2^+$, the reactions are



$$K_1 = \frac{[Ag(NH_3)^+]}{[Ag^+][NH_3]} = 2.1 \times 10^3 \quad K_2 = \frac{[Ag(NH_3)_2^+]}{[Ag(NH_3)^+][NH_3]} = 8.1 \times 10^3$$

$$K_f = K_1 K_2 = \frac{[Ag(NH_3)_2^+]}{[Ag^+][NH_3]^2} = 1.7 \times 10^7 \quad \text{at } 25^\circ C$$

Formation constant (or stability constant)





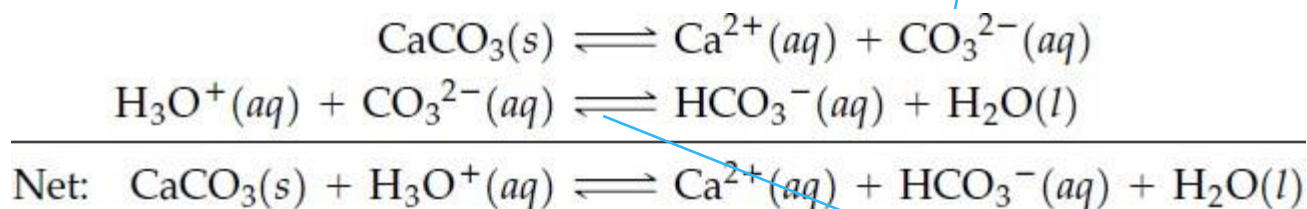
3. The pH of the Solution



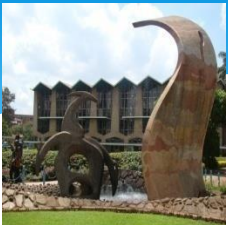
An ionic compound that contains a basic anion becomes more soluble as the acidity of the solution increases.

The solubility of CaCO_3 for example, increases with decreasing pH

As carbonate ions are removed from the solution, the solubility equilibrium shifts to the right.



The solubility of CaCO_3 at 25°C increases as the solution becomes more acidic because the carbonate ions combine with protons.



Quantitative Measurement of pH



- ✓ In case of acids, relative strength can be compared by measuring the concentration of hydrogen ions or by measuring their pH.
- ✓ The "p" in pH comes from German word "potenz" meaning power and "H" from $[H^+]$ concentration of hydrogen
- ✓ The pH of a solution is the negative logarithm to base 10 of the molar hydrogen ion concentration i.e. $pH = -\lg[H^+_{(aq)}]$

Notice that the accepted abbreviation for logarithm to base 10 is lg not log.

Some useful mathematical expressions

$$\lg xy = \lg x + \lg y$$

$$\lg \frac{x}{y} = \lg x - \lg y$$

$$\lg x^n = n \lg x$$

For example $\lg_{10} 2 = 0.3010$ implying $10^{0.3010} = 2$



Examples



1. What is the pH of $10^{-1} \text{ mol dm}^{-3}$ HCl (0.1 M HCl)?

Since HCl is fully dissociated and monobasic acid; $[H^+_{(aq)}]$ in $10^{-1} \text{ mol dm}^{-3}$ HCl = $10^{-1} \text{ mol dm}^{-3}$

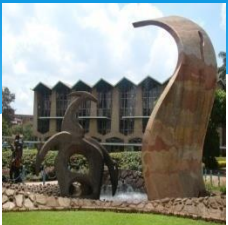
$$\begin{aligned} pH &= -\lg[H^+_{(aq)}] = -\lg 10^{-1} \\ &= (-1)\lg 10 \\ &= (-1)(-1) = +1 \end{aligned}$$

2. What is the pH of $10^{-3} \text{ mol dm}^{-3}$ H_2SO_4 (0.001 M H_2SO_4)? Since H_2SO_4 is fully dissociated and dibasic acid. $[H^+_{(aq)}]$ in $10^{-3} \text{ mol dm}^{-3}$ H_2SO_4 = $2 \times 10^{-3} \text{ mol dm}^{-3}$

$$\begin{aligned} pH &= -\lg[H^+_{(aq)}] = -\lg(2 \times 10^{-3}) \\ &= (-1)\lg 2 \times 10^{-3} \\ &= (-1)(0.30 - 3) = +2.7 \end{aligned}$$

3. What are the pH values of the following solutions?

- a) $10^{-3} \text{ mol dm}^{-3}$ HCl
- b) 1.0 mol dm^{-3} HCl
- c) 3 M HX which is only 50% dissociated



Buffer Solutions



Are solutions that resist change in pH on addition of acid or alkali.

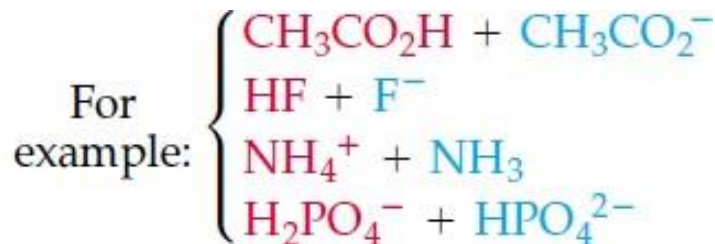
A buffer solution requires two components, one which is able to neutralize acids, and the other able to neutralize bases.

But of course the two components must not neutralize each other. Common buffer solutions are described either as a mixture of:

- *a weak acid and its conjugate base or*
- *a weak base and its conjugate acid.*

BUFFER SOLUTION

Weak acid
+
Conjugate base





The Henderson–Hasselbalch Equation



The H_3O^+ concentration in a buffer solution depends on the dissociation constant of the weak acid and on the concentration ratio [weak acid]/[conjugate base]:

$$[\text{H}_3\text{O}^+] = K_a \frac{[\text{Acid}]}{[\text{Base}]}$$

This equation can be rewritten in logarithmic form by taking the negative base-10 logarithm of both sides:

$$\text{pH} = -\log [\text{H}_3\text{O}^+] = -\log K_a - \log \frac{[\text{Acid}]}{[\text{Base}]}$$

If we then define $\text{p}K_a = -\log K_a$ by analogy with $\text{pH} = -\log [\text{H}_3\text{O}^+]$ and recognize that

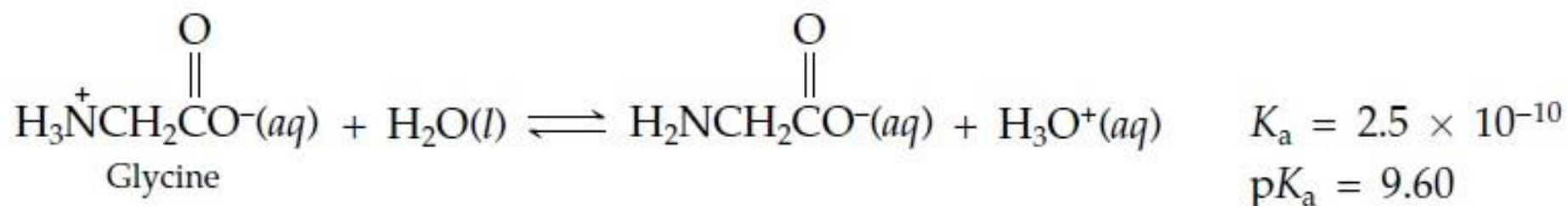
$$-\log \frac{[\text{Acid}]}{[\text{Base}]} = \log \frac{[\text{Base}]}{[\text{Acid}]}$$

we obtain an expression called the Henderson–Hasselbalch equation:

$$\text{HENDERSON–HASSELBALCH EQUATION} \quad \text{pH} = \text{p}K_a + \log \frac{[\text{Base}]}{[\text{Acid}]}$$



Suppose, for example, that you have a solution containing the amino acid glycine, one of the molecules from which proteins are made, and that the pH of the solution is 2.00 pH units greater than the pK_a of glycine:



$$\text{pH} = pK_a + 2.00 = 11.60$$

Since $\text{pH} = pK_a + 2.00$, then $\log [\text{base}]/[\text{acid}] = 2.00$, and $[\text{base}]/[\text{acid}] = 1.0 \times 10^2 = 100/1$. According to the Henderson–Hasselbalch equation, therefore, 100 of every 101 glycine molecules are dissociated, which corresponds to 99% dissociation:

$$\log \frac{[\text{Base}]}{[\text{Acid}]} = \text{pH} - pK_a = 2.00$$

$$\frac{[\text{Base}]}{[\text{Acid}]} = 1.0 \times 10^2 = \frac{100}{1} \quad 99\% \text{ dissociation}$$



What is the pH a buffer solution made from 0.1 M acetic acid and 0.1 M sodium acetate solution, given K_a (acetic acid) = 1.8×10^{-5} M)?



Solution

$$pH = pK_a + \lg \frac{[conjugatebase]}{[acid]}$$

$$pH = -\lg (1.8 \times 10^{-5}) K_a + \lg \frac{0.10}{0.10}$$

4.74

END OF LECTURE 5 and 6